

Topological 0-Matrix_(Substrate) Mechanics

This monograph presents a complete, alternative foundation for fundamental physics, achieving a radical simplification of the description of reality by eliminating the abstract concept of empty spacetime in favor of continuum mechanics. **Topological 0-Matrix (Substrate) Mechanics (TSM)** proves that the entire observable Universe emerges as a macroscopic, elasto-dynamic manifestation of a sub-Planckian, geometrically jammed network of oscillators—the titular 0-Matrix.

The TSM theory fulfills the grand dream of theoretical physics for complete unification, delivering unprecedented simplifications in two key areas:

- **Derivation of General Relativity (GR):** Gravity is not the magic of curved spacetime invented by Einstein. It is directly derived from classical, **4-dimensional engineering Navier-Cauchy equations!** Space is a tangible medium, its metric is the state of internal stresses, and attraction is a mere pressure gradient. **TSM uncompromisingly eliminates singularities (no black holes)** and, through a single, fundamental elasto-dynamic mechanism, **erases the mythical dark matter and dark energy from physics.**
 - **Simplification and Demystification of the Standard Model:** No more absurd, dimensionless material points and quantum mysticism. In TSM, elementary particles are reduced to stable, non-linear topological knots (solitons) within the 0-Matrix substrate. Rest mass, spin 1/2, and charge are pure geometry of standing waves trapped in the network. Consequently, the model explains the internal structure of fermions with rigorous, mechanical precision. **It explains and derives the puzzling mass relations of particles, including the famous Koide formula.**
-

Table of Contents

- Chapter 0: Axioms of the 0-Matrix, or the Mechanical "Engine" of Reality
 - 0.1. Rejection of the Non-Physical Void
 - 0.2. Elasticity from Pure Kinetics
 - 0.3. The 4D Pressure Vessel: Finiteness as a Source of Stress
 - 0.4. Relaxation State: Autoregulation of 0-Particles
 - 0.5. Birth of the Membrane (3D Self-Organization)
 - 0.6. Axiom of Physical Finiteness: Categorical Exclusion of Singularities and "Black Holes"
 - 0.7. Creation of the First Distortions: An Alternative to the Big Bang
 - Chapter 1: Axiomatics and Rigorous Formalism of TSM: Emergent Time, Chaotic Substrate, and Elasto-Dynamics of the 3-Brane
 - ► **1.1. Fundamental Evolutionary Parameter τ as a Measure of Global, Averaged 0-Particle Activity**
 - ► **1.2. Medium and Space: The Geometric Jamming State and the Isotropic 3-Brane**
-

- 1.3. Non-linear Kinematics: Emergence of the Metric g_{ab} from Strain
- 1.4. Wave Dynamics of the Substrate and Field Equations
- ► **1.5. Dual Regime of Elasto-Dynamics: Elastic Waves vs. Non-linear Topological Locking**
- ► **1.6. Axiom of Matter: Quantized Topological Knots and Rigorous Separation of Mass from Charge**
- ► **1.7. Derivation of Relativistic Kinematics from the Hydro-Elasticity of the Substrate**
- 1.8. Variable Wave Propagation (Local $c_{\perp}$) and Material Constants
- 1.9. Summary of Chapter 1

- Chapter 2: Topological Knots and Solitonic Emergence in TSM: Hydrodynamics of Inertia, Antimatter, Geometric Origin of Spin 1/2, and the Substrate Stiffness Formalism
 - ► **2.1. Microscopic Mechanism of Finite Compressibility and Rigorous Elasto-Dynamic Formalism in $\mathbb{R}^4$**
 - 2.2. Breakdown of Superposition, Mode Conversion, and Topological Knots in the Crystalline 3-Brane
 - 2.3. Hydrodynamic Mechanism of Inertia and the Peierls-Nabarro Barrier
 - 2.4. The Essence of Charge and Separation from Rest Mass
 - 2.5. Antimatter as Topological Inversion
 - 2.6. Geometric Origin of Spin 1/2 and Double Covering
 - 2.7. Summary of Chapter 2

- Chapter 3: Stress Dynamics of the 0-Matrix: From the Unification of Electrodynamics and Nuclear Forces to Circuit Mechanics
 - 3.1. From Topological Torsion to Gauge Potential
 - 3.2. Shear Stress Tensor vs. Electromagnetic Field Tensor
 - 3.3. Conservation Laws and Wave Dynamics
 - ► **3.4. Interaction Mechanics: Non-linear Field Equations, Feedbacks, and Maxwell Forces**
 - ► **3.5. Ontology of the Strong Interaction: Non-linear Coupling of Topological Torsions at the Femtometer Scale**
 - 3.6. Voltage and Current as Wave Mechanics in Conductors
 - 3.7. Electromechanical Inertia: Reactance and Phase Shifts
 - 3.8. Quantum Applications: Superconductivity and the Hall Effect
 - 3.9. Summary of Chapter 3

- Chapter 4: Thermodynamics of the Substrate: Stochastic Noise, Spectral Entropy, and Non-linear Relaxation of the 0-Matrix
 - ► **4.1. Physical Nature and Temperature Dualism in TSM**
 - ► **4.2. Ontology and Time Dualism in TSM: Statistical Emergence of Parameter τ and Emergence of Time t**
 - 4.3. Energy Dissipation Mechanism: Decay of Wave Coherence into 0-Matrix Granularity
 - ► **4.4. Entropy as Informational Vibration Chaos (Statistical Description of the 0-Matrix)**
 - 4.5. Arrow of Time: Direction of Irreversible Energy Dissipation
 - 4.6. Thermodynamic Equilibrium in Open Systems (Energy Exchange with 4D)
 - ► **4.7. Empirical Verification, Theoretical Gaps, and Potential Falsifications**
 - 4.8. Summary of Chapter 4
-

- Chapter 5: Topological Exclusion Mechanics and Stability of Matter
 - 5.1. Classification of Matrix Excitations: Structural Knots vs. Wave Modes
 - 5.2. Mathematical Formalism of Topological Exclusion (Pauli)
 - 5.3. Contrast with Bosons: Bose-Einstein Condensation (BEC) as Wave Interference
 - 5.4. Degenerate Matter Pressure and the Cosmological Problem of Type Ia Supernovae
 - ► **5.5. Observational Verification of the Topological Model**
 - 5.6. Summary of Chapter 5
-

- Chapter 6: Gravity as Orthogonal Tension: 4D Dimension, Plasmoids, and Non-linear Continuum Dynamics
 - 6.1. Ontology of Gravity and the Linear Limit (Laplace's Equation)
 - 6.2. Formalized Elasto-Dynamics of the 3-Brane: Non-linear Deflection in $\mathbb{R}^4$
 - 6.3. Topological Plasmoids Instead of Black Holes
 - ► **6.4. Galactic Dynamics: Torsional Dragging and Elimination of Dark Matter**
 - ► **6.5. Alternative to Expansion: Redshift as Distance-Based Wave Dispersion (Consistency with Postulate 0.7)**
 - ► **6.6. Physical Nature of the Constant c and Stress Waves (LIGO)**
 - 6.7. Summary of Chapter 6
-

- Chapter 7: Subatomic Interactions, Hadron Physics, and Quantum Determinism in Topological Geometrodynamics of the Matrix (TGM)

- 7.1. What is the Strong Interaction in the Standard Framework?
- 7.2. Cavitation Model of the Proton vs. Deep Inelastic Scattering (DIS)
- 7.3. Topological Confinement
- 7.4. Asymptotic Freedom
- 7.5. Hadron Masses (Proton, Neutron)
- 7.6. Absence of Free Gluons
- ► **7.7. Internal Quark Dynamics: Running Masses, Hierarchy, and the Physical Meaning of the CKM Matrix**
- 7.8. Reinterpretation of Heavy Particles: Topo-Harmonic Excitations of the Matrix
- 7.9. Demystification of Quantum Mechanics: Young's Experiment and the Observer Effect as Hydrodynamic Phenomena
- 7.10. Matrix Resonator – How to Excite Controlled Vibrations of the 0-Matrix
- 7.11. Accelerator Tests – Falsification Proposals
- 7.12. Topological Test – Searching for $\mathfrak{Z}_1$ Knot Rupture Patterns in LHC Data
- 7.13. Summary of Chapter 7
- 7.14. Empirical Foundation and Data Sources

- Chapter 8: Weak Interactions and Topological Lepton Hierarchy: Stochastic Degradation, Phase Lock, and $SU(2)$ Geometry
 - 8.1. Free Neutron as a Stochastically Decaying Oscillator
 - 8.2. Phase Lock: Kinematics of Atomic Nuclei Stability
 - 8.3. Neutron Lifetime Anomaly as Verification of the Phase Lock
 - 8.4. W and Z Resonances as Topo-Harmonic Excitations (Breathers)
 - 8.5. Width of the Z Resonance vs. Topological Isotropy of the $SU(2)$ Algebra
 - ► **8.6. Topological Hierarchy of Lepton Masses: Muon and Tauon as Higher-Order Link Excitations**
 - 8.7. Summary of Chapter 8
 - 8.8. Empirical Foundation and Data Sources

- Chapter 9: The Illusion of Dark Matter and Dark Energy in TGM: Elasto-Dynamics of Rotation, Non-linear Lensing, and the CMB

- 9.1. The Dark Matter Problem in Standard Astrophysics
 - ► **9.2. Elasto-Dynamics of Galaxies: Stress Halo and Background Pressure Gradient**
 - ► **9.3. Non-linear Lensing in Clusters: Stress Synergy**
 - 9.4. Apparent Cosmic Expansion as Stress Relaxation (The Illusion of Dark Energy)
 - 9.5. Cosmic Microwave Background (CMB) as Thermo-Acoustic Matrix Noise
 - 9.6. Comparison with MOND and the NFW Profile
 - 9.7. Experimental Tests and Falsifiability of TGM in Astrophysics
 - 9.8. Summary of Chapter 9
-

- Chapter 10: Central Plasmoids and Torsional Electrodynamics: The Illusion of Black Holes and Refractive Shadows
 - 10.1. The GR Singularity Problem and the Algorithmic Error of the EHT Telescope
 - 10.2. The Central Plasmoid as a Compact State of the Degenerate 0-Matrix
 - 10.3. Black Hole Shadow as a Spherical Refractive Trap (The Illusion of V_{ij})
 - 10.4. Torsional Electrodynamics of the Core: Z-pinch and Matrix Vortices
 - 10.5. Galactic Jets as Pressure Valves and Matrix Cavitation
 - ► **10.6. Birth of Quasars at Jet Terminations: The Arp Mechanism and Intrinsic Redshift Conditions**
 - 10.7. Summary of Chapter 10
-

- Chapter 11: Non-Expanding Cosmology – Matrix Relaxation, SN Ia Time Dilation, and Multi-layered Wave Propagation
 - 11.1. Rejection of Cosmic Expansion
 - 11.2. 0-Matrix Relaxation – Cosmological Evolution of Mean Stress
 - ► **11.3. Multi-layered Light Propagation – Three Stress Zones**
 - 11.4. Intrinsic vs. Cosmological Redshift – Rehabilitation of Halton Arp
 - 11.5. Tired Light as a Second-Order Effect (Scattering on 0-Particles)
 - 11.6. Tests and Predictions
 - 11.7. Summary of Chapter 11
-

- Chapter 12: Experimental Verification Program – Falsifiability, Resonant Test, and Demystification of Quantum Entanglement

- ► **12.1. Priority Test: Variation of the Speed of Light in a Magnetic Field – Resonant Variant**
- 12.2. Full Compendium of Crucial Falsification Tests
- ► **12.3. Quantum Entanglement vs. Bell's Tests – Limits of Non-locality**
- 12.4. Summary of Chapter 12

- Chapter 13: Ontological Summary, Fundamental Constants, and the Ultimate Unification of Physics
 - 13.1. Compendium of the Theory's Axioms
 - 13.2. Fundamental Material Parameters of the Universe
 - 13.3. Ultimate Unification – TGM Model vs. Classical Physics
 - 13.4. Open Engineering Challenges
 - 13.5. Concluding Words

Methodological Genesis and Formal Framework of TSM Unification

This treatise presents a proposal for the unification of General Relativity (GR) and the Standard Model (SM). Grounded in data accumulated through decades of observations and experimental research, Topological Substrate Mechanics (TSM) merges these two pillars of modern physics into a single, coherent framework. The formulation of this mathematical formalism would not have been possible without building upon the groundbreaking work of numerous outstanding theoretical physicists and mathematicians. TSM synthesizes their key achievements, reshaping the existing, rigorous conceptual framework into a unified language for describing reality that transcends current divisions.

The scientific fields and achievements from whose apparatus and conceptual heritage TSM draws represent well-established pillars of modern mathematical physics and theoretical mechanics. This theory does not introduce postulates *ex nihilo*, but rather performs a profound synthesis and paradigmatic reinterpretation of mathematical structures developed within the following disciplines:

1. Continuum mechanics and condensed matter physics Introducing the classical formalism of stress-strain tensors (the foundations of which were laid by Cauchy and Navier in the **1820s and 1830s**) and nonlinear elastodynamics into TSM allows for the formal description of the Topological Substrate as a real, continuous physical medium. This apparatus, further developed in the **mid-20th century** for condensed matter physics (notably by the Landau school), characterizes the medium using macroscopic parameters such as the bulk modulus and geometric packing density. This approach replaces the abstract, material-substrate-free spacetime background of GR with the mechanistic concept of material resistance and substrate elasticity.

2. Soliton physics and nonlinear differential equations The utilization of the mathematical apparatus of soliton solutions forms the bedrock of the theory of matter structure in TSM. While the observation of the solitary wave itself dates back to 1834 (J. S. Russell), the mathematical renaissance of this field occurred in the **1960s and 1970s** with the formulation of the nonlinear Schrödinger equation and the Sine-Gordon equation. Stable, non-local, and non-dispersive waves (solitons) provide a rigorous mathematical description of elementary particles, eliminating the inherently contradictory concept of a point particle.

3. Knot theory and algebraic topology The application of topological invariants and braid theory provides formal tools for categorizing charges and particles as geometric braids. This discipline, initiated by Henri Poincaré at the end of the 19th century, experienced its crucial expansion for physics in the **1980s** (largely due

to the work of Vaughan Jones and Edward Witten). Algebraic topology guarantees the absolute permanence of material structures in TSM, explaining conservation laws through the geometric impossibility of a knot spontaneously untying itself within a continuum.

4. Mechanics of lattice defects and dislocation models The adaptation of microstructural models of dislocations in crystals relies directly on the **Peierls-Nabarro model (formulated in 1940 and modified in 1947)**. The implementation of these advanced nonlinear equations allows for the description of the kinetics of solitons moving through the Substrate lattice. This enables the mechanistic derivation of resistance to defect motion (so-called lattice friction), which finds direct application in modeling astrophysical processes, such as the nonlinear generation of plasma jets.

5. Kinetic theory of gases and statistical mechanics of jammed systems TSM combines classic Boltzmann transport equations (**1872**) with the modern theory of densely packed systems and the jamming transition, which was defined in complex systems physics in the late **1990s (specifically in 1998 by Liu and Nagel)**. This allows for the microscopic derivation of the macroscopic elastic properties of the Substrate from the chaos of sub-Planckian collisions of primal elements (0-particles), forming the mathematical core of the TSM Equation of State (TSM-EOS).

6. Differential geometry and multidimensional membrane models In terms of multidimensional description, TSM draws on the evolution of Riemannian geometry (since **1854**), Kaluza-Klein theory (the **1920s**), up to the Randall-Sundrum braneworld cosmology models of **1999**. However, the theory translates these structures from the language of abstract geometry into the language of pressure physics. The three-dimensional Membrane (our Universe) is defined as a physical phase boundary of finite thickness, subject to warping into the fourth spatial dimension under the influence of local stress gradients inside a pressurized vessel (the Bulk).